

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[NG_FFCutsFits,CutFitFileNG] = recover_selected_cuts('NG_FFCutsFits');
```

```
fn = fieldnames(NG_FFCutsFits);
nf = 11
```

```
nf =
11
```

```
%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei200_off200NGmaxFF'
```

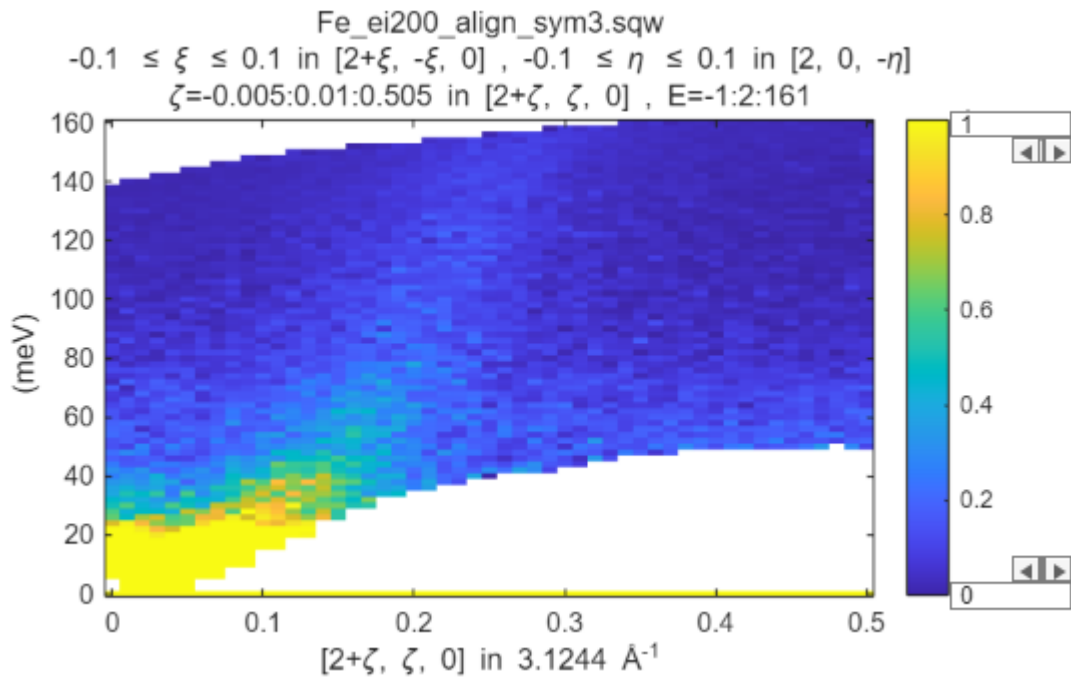
```
selectCuts =
'Ei200_off200NGmaxFF'
```

```
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end
```

```
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei200_off200NGmaxFF
```

```
cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
    cuts_list{j} = cuts_list{j}.set_instrument(mi);
    cuts_list{j} = cuts_list{j}.set_sample(sample);
    plot(cuts_list{j}); lz 0 1; keep_figure;
end
```



```
dir_name = "NG";

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 380;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;

pNG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

**** En = 35±5 going to 375

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.84264s Reduced Chi^2 = 33.1708

Free parameter values:

10	0.7917	34	0.5293
----	--------	----	--------

Iteration = 1

Total time = 3.0058s Reduced Chi^2 = 13.3744 Levenberg-Marquardt = 0.1

Free parameter values:

12.18	0.2993	62.76	0.3858
-------	--------	-------	--------

Iteration = 2

```

-----
Total time = 5.1256s    Reduced Chi^2 = 26.9138    Levenberg-Marquardt = 0.01
Free parameter values:
    9.384          1.412          76.99          0.2297
Total time = 5.6797s    Reduced Chi^2 = 30.3377    Levenberg-Marquardt = 0.1
Free parameter values:
    7.555          1.182          65.49          0.2724
Total time = 6.2164s    Reduced Chi^2 = 13.4213    Levenberg-Marquardt = 1
Free parameter values:
    8.834          0.7576          68.82          0.3464
Total time = 6.7445s    Reduced Chi^2 = 11.5005    Levenberg-Marquardt = 100
Free parameter values:
    11.63          0.3575          64.47          0.3835
-----
Iteration = 3
-----
Total time = 8.883s    Reduced Chi^2 = 8.1618    Levenberg-Marquardt = 10
Free parameter values:
    10.98          0.525          64.39          0.3739
-----
Iteration = 4
-----
Total time = 10.9995s    Reduced Chi^2 = 4.2474    Levenberg-Marquardt = 1
Free parameter values:
    10.86          0.8764          58.8          0.3325
-----
Iteration = 5
-----
Total time = 13.1147s    Reduced Chi^2 = 1.8388    Levenberg-Marquardt = 0.1
Free parameter values:
    10.37          1.308          59.02          0.2509
-----
Iteration = 6
-----
Total time = 15.1763s    Reduced Chi^2 = 1.6479    Levenberg-Marquardt = 0.01
Free parameter values:
    10.91          1.457          61.05          0.2211
-----
Iteration = 7
-----
Total time = 17.363s    Reduced Chi^2 = 1.6304    Levenberg-Marquardt = 0.001
Free parameter values:
    12.04          1.523          68.06          0.2104
-----
Iteration = 8
-----
Total time = 19.461s    Reduced Chi^2 = 1.5962    Levenberg-Marquardt = 0.0001
Free parameter values:
    12.44          1.546          70.69          0.2071
-----
Iteration = 9
-----
Total time = 21.6296s    Reduced Chi^2 = 1.6325    Levenberg-Marquardt = 1e-05
Free parameter values:
    12.52          1.545          71.25          0.2076
Total time = 22.1625s    Reduced Chi^2 = 1.6129    Levenberg-Marquardt = 0.0001
Free parameter values:
    12.52          1.545          71.22          0.2077
Total time = 22.6975s    Reduced Chi^2 = 1.6024    Levenberg-Marquardt = 0.001
Free parameter values:
    12.49          1.544          71.05          0.2079
Total time = 23.2416s    Reduced Chi^2 = 1.6194    Levenberg-Marquardt = 0.1
Free parameter values:
    12.45          1.543          70.76          0.2079
Total time = 23.7604s    Reduced Chi^2 = 1.6027    Levenberg-Marquardt = 10
Free parameter values:
    12.44          1.545          70.7          0.2073
Total time = 24.2847s    Reduced Chi^2 = 1.596    Levenberg-Marquardt = 1000

```

Free parameter values:
12.44 1.546 70.69 0.2071

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	12.44	+/-	6.525
4	1.546	+/-	0.3379
5	0		
6	70.69	+/-	41.94
7	0		
8	0		
9	0		
10	0		

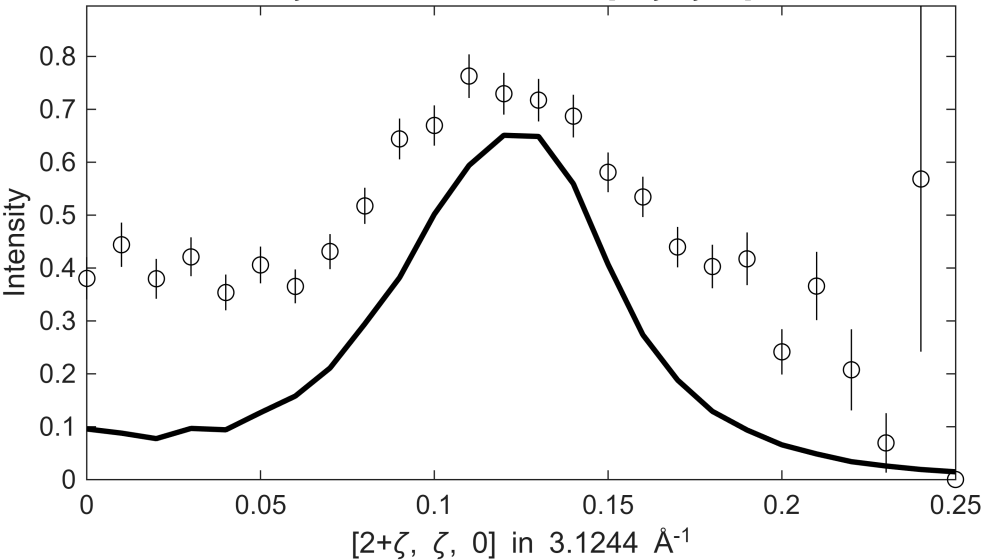
Background parameter values:

1	0.2071	+/-	0.05402
2	0		
3	0		

Covariance matrix for free parameters:

1.0000	0.9470	0.9993	-0.9141
0.9470	1.0000	0.9457	-0.9792
0.9993	0.9457	1.0000	-0.9150
-0.9141	-0.9792	-0.9150	1.0000

Fe_ei200_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2+ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, -η] , 30 ≤ E ≤ 40
ζ=-0.005:0.01:0.505 in [2+ζ, ζ, 0]



*** En = 45±5 going to 375

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Beginning fit (max 20 iterations)-----
Starting point

Total time = 0.73655s Reduced Chi^2 = 58.3614

```

Free parameter values:
    12.44      1.546      70.69      0.3044
-----
Iteration = 1
-----
Total time = 3.763s    Reduced Chi^2 = 1.9621    Levenberg-Marquardt = 0.1
Free parameter values:
    13.82      0.9874      69.32      0.1662
-----
Iteration = 2
-----
Total time = 6.887s    Reduced Chi^2 = 1.1828    Levenberg-Marquardt = 0.01
Free parameter values:
    14.6       1.091      65.98      0.1497
-----
Iteration = 3
-----
Total time = 9.9641s   Reduced Chi^2 = 1.1638    Levenberg-Marquardt = 0.001
Free parameter values:
    14.57      1.123      66.16      0.1467
-----
Iteration = 4
-----
Total time = 12.9962s  Reduced Chi^2 = 1.1606    Levenberg-Marquardt = 0.0001
Free parameter values:
    14.51      1.12      65.93      0.1469
-----
Iteration = 5
-----
Total time = 16.0177s  Reduced Chi^2 = 1.1661    Levenberg-Marquardt = 1e-05
Free parameter values:
    14.32      1.118      64.95      0.1471
Total time = 16.7962s  Reduced Chi^2 = 1.1628    Levenberg-Marquardt = 0.0001
Free parameter values:
    14.32      1.118      64.96      0.1471
Total time = 17.5926s  Reduced Chi^2 = 1.1691    Levenberg-Marquardt = 0.001
Free parameter values:
    14.34      1.118      65.08      0.1471
Total time = 18.3506s  Reduced Chi^2 = 1.1617    Levenberg-Marquardt = 0.1
Free parameter values:
    14.48      1.121      65.76      0.1468
Total time = 19.1124s  Reduced Chi^2 = 1.1717    Levenberg-Marquardt = 10
Free parameter values:
    14.51      1.12      65.92      0.1469
Total time = 19.8986s  Reduced Chi^2 = 1.161     Levenberg-Marquardt = 1000
Free parameter values:
    14.51      1.12      65.93      0.1469
*** No improvement in chi-squared over previous iteration ***
Total time = 19.9059s  Reduced Chi^2 = 1.1606
Free parameter values:
    14.51      1.12      65.93      0.1469
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

Foreground parameter values:
-----
  1          1
  2          8
  3      14.51 +/- 0.2595
  4       1.12 +/- 0.05302
  5          0
  6      65.93 +/- 1.716
  7          0
  8          0
  9          0

```

Background parameter values:

1	0.1469	+/-	0.007391
2	0		
3	0		

Covariance matrix for free parameters:

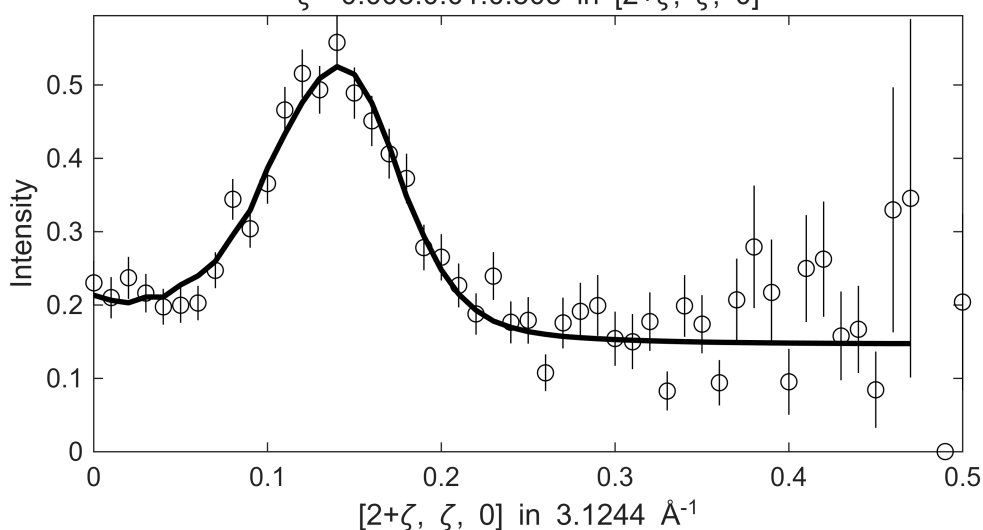
1.0000	-0.1984	0.5891	-0.4747
-0.1984	1.0000	-0.3729	-0.6773
0.5891	-0.3729	1.0000	-0.3749
-0.4747	-0.6773	-0.3749	1.0000

Fe_ei200_align_sym3.sqw

S= 1.1±0.053; J0= 66± 1.7; J1= 0± 0; gamma=14.514±0.259514

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, -η] , 40 ≤ E ≤ 50

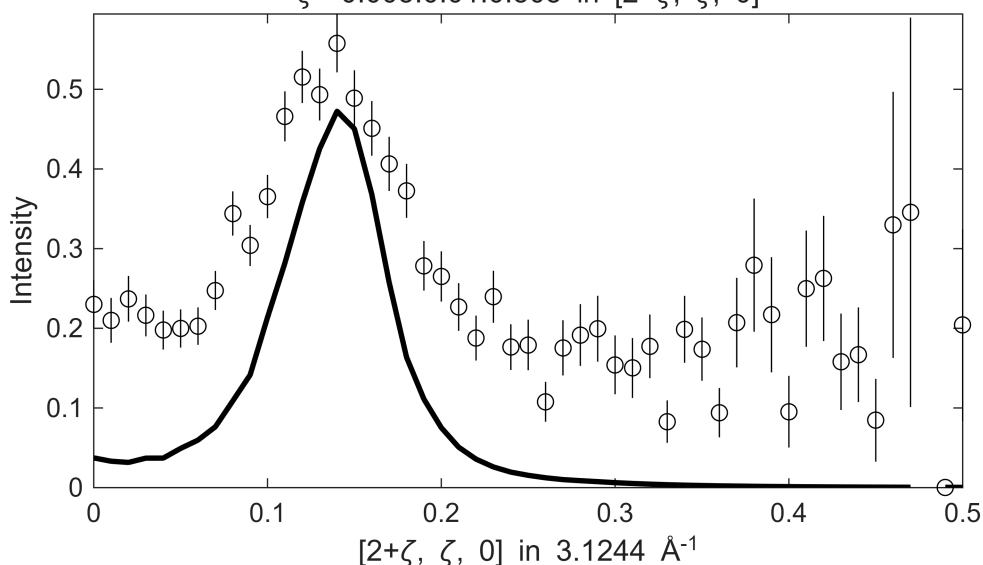
ζ=-0.005:0.01:0.505 in [2+ζ, ζ, 0]



Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, -η] , 40 ≤ E ≤ 50

ζ=-0.005:0.01:0.505 in [2+ζ, ζ, 0]



 **** En = 55±5 going to 375

Beginning fit (max 20 iterations)-----

Starting point

Total time = 1.5314s Reduced Chi^2 = 25.2782
Free parameter values:
14.51 1.12 65.93 0.2036

Iteration = 1

Total time = 7.4754s Reduced Chi^2 = 1.2269 Levenberg-Marquardt = 0.1
Free parameter values:
15.71 0.6501 63.53 0.1574

Iteration = 2

Total time = 13.5773s Reduced Chi^2 = 0.87788 Levenberg-Marquardt = 0.01
Free parameter values:
13.17 0.6536 45.48 0.155

Iteration = 3

Total time = 19.6489s Reduced Chi^2 = 0.75783 Levenberg-Marquardt = 0.001
Free parameter values:
8.848 0.6074 32.32 0.1571

Iteration = 4

Total time = 25.6133s Reduced Chi^2 = 0.72986 Levenberg-Marquardt = 0.0001
Free parameter values:
9.336 0.6187 33.35 0.1568

Iteration = 5

Total time = 31.5664s Reduced Chi^2 = 0.72762 Levenberg-Marquardt = 1e-05
Free parameter values:
9.016 0.6146 32.18 0.157

Iteration = 6

Total time = 37.4969s Reduced Chi^2 = 0.73321 Levenberg-Marquardt = 1e-06
Free parameter values:
8.974 0.6152 32.03 0.157
Total time = 39.0049s Reduced Chi^2 = 0.73279 Levenberg-Marquardt = 1e-05
Free parameter values:
8.974 0.6152 32.03 0.157
Total time = 40.5143s Reduced Chi^2 = 0.73344 Levenberg-Marquardt = 0.0001
Free parameter values:
8.974 0.6152 32.04 0.157
Total time = 42.0254s Reduced Chi^2 = 0.73585 Levenberg-Marquardt = 0.01
Free parameter values:
8.997 0.6155 32.12 0.157
Total time = 43.546s Reduced Chi^2 = 0.7374 Levenberg-Marquardt = 1
Free parameter values:
9.013 0.6153 32.18 0.157
Total time = 45.0166s Reduced Chi^2 = 0.73505 Levenberg-Marquardt = 100
Free parameter values:
9.016 0.6147 32.18 0.157
*** No improvement in chi-squared over previous iteration ***
Total time = 45.0239s Reduced Chi^2 = 0.72762
Free parameter values:
9.016 0.6146 32.18 0.157

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	9.016	+/-	0.2081
4	0.6146	+/-	0.03135
5	0		
6	32.18	+/-	0.9059
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.157	+/-	0.003003
2	0		
3	0		

Covariance matrix for free parameters:

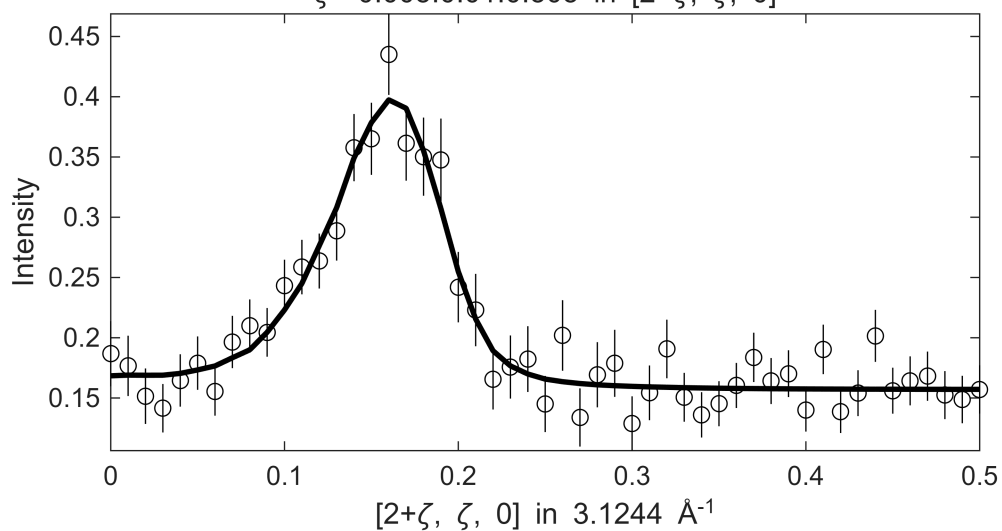
1.0000	-0.8435	0.6996	0.0364
-0.8435	1.0000	-0.8651	-0.3913
0.6996	-0.8651	1.0000	0.0006
0.0364	-0.3913	0.0006	1.0000

Fe_ei200_align_sym3.sqw

S=0.61±0.031; J0= 32±0.91; J1= 0± 0; gamma=9.01629±0.208112

-0.1 ≤ ξ ≤ 0.1 in [2+ ξ , - ξ , 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, - η] , 50 ≤ E ≤ 60

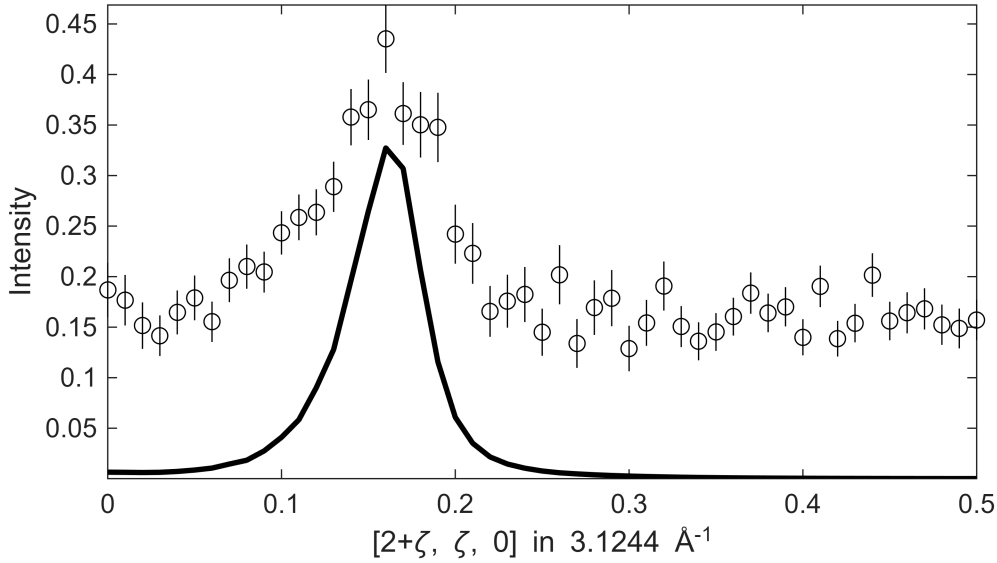
ζ =-0.005:0.01:0.505 in [2+ ζ , ζ , 0]



Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $50 \leq E \leq 60$

$\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 65±5 going to 375

Beginning fit (max 20 iterations)-----

Starting point

Total time = 1.6703s Reduced Chi^2 = 9.3812

Free parameter values:

9.016 0.6146 32.18 0.1669

Iteration = 1

Total time = 7.9968s Reduced Chi^2 = 1.6236 Levenberg-Marquardt = 0.1

Free parameter values:

10.52 0.5966 35.91 0.1215

Iteration = 2

Total time = 14.3186s Reduced Chi^2 = 1.3906 Levenberg-Marquardt = 0.01

Free parameter values:

14.7 0.6708 49.58 0.1161

Iteration = 3

Total time = 20.6681s Reduced Chi^2 = 1.2726 Levenberg-Marquardt = 0.001

Free parameter values:

20.97 0.7489 71.14 0.1124

Iteration = 4

Total time = 27.073s Reduced Chi^2 = 1.2617 Levenberg-Marquardt = 0.0001

Free parameter values:

24.37 0.7816 83.82 0.1109

Iteration = 5

Total time = 33.383s Reduced Chi^2 = 1.2633 Levenberg-Marquardt = 1e-05

Free parameter values:

25.12 0.787 86.81 0.1107

Total time = 34.9773s Reduced Chi^2 = 1.2601 Levenberg-Marquardt = 0.0001

Free parameter values:

25.11 0.787 86.78 0.1107

Iteration = 6

```

-----
Total time = 41.3645s    Reduced Chi^2 = 1.2541    Levenberg-Marquardt = 1e-05
Free parameter values:
      25.2      0.7882      87.08      0.1107
-----

```

Iteration = 7

```

-----
Total time = 47.7651s    Reduced Chi^2 = 1.2595    Levenberg-Marquardt = 1e-06
Free parameter values:
      25.14      0.7868      86.92      0.1107
Total time = 49.3869s    Reduced Chi^2 = 1.2608    Levenberg-Marquardt = 1e-05
Free parameter values:
      25.14      0.7868      86.92      0.1107
Total time = 51.005s     Reduced Chi^2 = 1.2581    Levenberg-Marquardt = 0.0001
Free parameter values:
      25.14      0.7868      86.92      0.1107
Total time = 52.657s     Reduced Chi^2 = 1.2588    Levenberg-Marquardt = 0.01
Free parameter values:
      25.16      0.787      87.01      0.1107
Total time = 54.2673s    Reduced Chi^2 = 1.2659    Levenberg-Marquardt = 1
Free parameter values:
      25.19      0.7876      87.09      0.1107
Total time = 55.8789s    Reduced Chi^2 = 1.2625    Levenberg-Marquardt = 100
Free parameter values:
      25.2      0.7881      87.08      0.1107
*** No improvement in chi-squared over previous iteration ***
Total time = 55.886s     Reduced Chi^2 = 1.2541
Free parameter values:
      25.2      0.7882      87.08      0.1107
-----

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
1          1
2          8
3          25.2 +/- 0.5417
4          0.7882 +/- 0.04586
5          0
6          87.08 +/- 3.107
7          0
8          0
9          0
10         0

```

Background parameter values:

```

-----
1          0.1107 +/- 0.00356
2          0
3          0

```

Covariance matrix for free parameters:

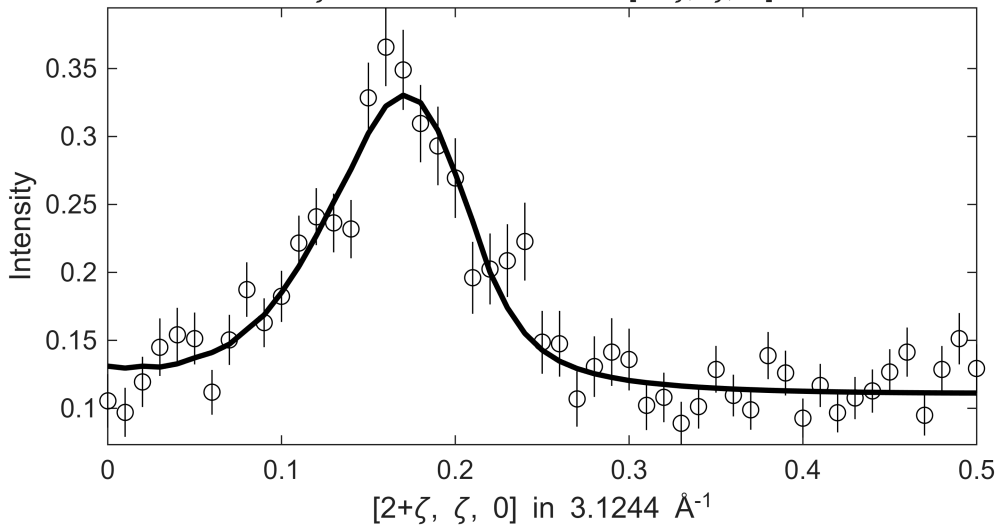
```

-----
1.0000 -0.6376 0.5625 -0.1301
-0.6376 1.0000 -0.8042 -0.4855
0.5625 -0.8042 1.0000 -0.0369
-0.1301 -0.4855 -0.0369 1.0000

```

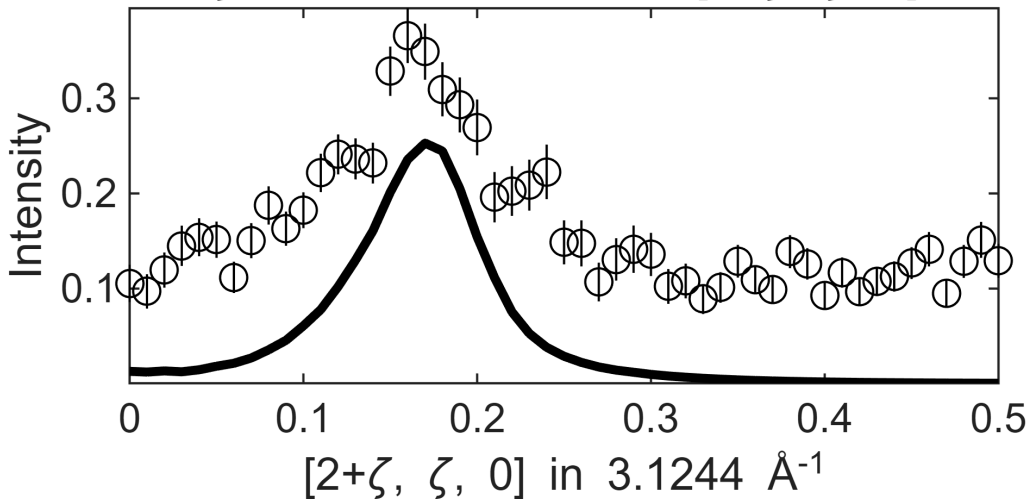
Fe_ei200_align_sym3.sqw

S=0.79±0.046; J0= 87± 3.1; J1= 0± 0; gamma=25.1965±0.541746
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $60 \leq E \leq 70$
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, ξ
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



*** En = 75±5 going to 375

Beginning fit (max 20 iterations)-----

Starting point

Total time = 1.7098s Reduced Chi^2 = 15.9421

Free parameter values:

25.2 0.7882 87.08 0.1257

Iteration = 1

Total time = 8.476s Reduced Chi^2 = 0.96362 Levenberg-Marquardt = 0.1

Free parameter values:

26.67 0.6539 82.98 0.08211

Iteration = 2

Total time = 15.2049s Reduced Chi^2 = 0.86085 Levenberg-Marquardt = 0.01

Free parameter values:

24.81	0.6807	73.33	0.07899
-------	--------	-------	---------

Iteration = 3

Total time = 21.9778s	Reduced Chi^2 = 0.85738	Levenberg-Marquardt = 0.001
Free parameter values:		
23.15	0.6701	68.43
		0.07955

Iteration = 4

Total time = 28.6846s	Reduced Chi^2 = 0.8576	Levenberg-Marquardt = 0.0001
Free parameter values:		
23.24	0.6706	68.76
		0.07952
Total time = 30.3916s	Reduced Chi^2 = 0.85755	Levenberg-Marquardt = 0.001
Free parameter values:		
23.23	0.6705	68.73
		0.07952
Total time = 32.0766s	Reduced Chi^2 = 0.85788	Levenberg-Marquardt = 0.01
Free parameter values:		
23.2	0.6703	68.63
		0.07954
Total time = 33.749s	Reduced Chi^2 = 0.85481	Levenberg-Marquardt = 1
Free parameter values:		
23.15	0.67	68.46
		0.07955

Iteration = 5

Total time = 40.6134s	Reduced Chi^2 = 0.85913	Levenberg-Marquardt = 0.1
Free parameter values:		
23.15	0.6699	68.44
		0.07954
Total time = 42.2655s	Reduced Chi^2 = 0.85756	Levenberg-Marquardt = 1
Free parameter values:		
23.15	0.67	68.45
		0.07954
Total time = 43.949s	Reduced Chi^2 = 0.85382	Levenberg-Marquardt = 10
Free parameter values:		
23.15	0.67	68.46
		0.07955

Iteration = 6

Total time = 50.7845s	Reduced Chi^2 = 0.8626	Levenberg-Marquardt = 1
Free parameter values:		
23.15	0.6699	68.46
		0.07955
Total time = 52.4739s	Reduced Chi^2 = 0.85627	Levenberg-Marquardt = 10
Free parameter values:		
23.15	0.6699	68.46
		0.07955
Total time = 54.1874s	Reduced Chi^2 = 0.85827	Levenberg-Marquardt = 100
Free parameter values:		
23.15	0.67	68.46
		0.07955
Total time = 55.9088s	Reduced Chi^2 = 0.85793	Levenberg-Marquardt = 10000
Free parameter values:		
23.15	0.67	68.46
		0.07955
Total time = 57.5936s	Reduced Chi^2 = 0.85954	Levenberg-Marquardt = 1000000
Free parameter values:		
23.15	0.67	68.46
		0.07955
Total time = 59.279s	Reduced Chi^2 = 0.85801	Levenberg-Marquardt = 100000000
Free parameter values:		
23.15	0.67	68.46
		0.07955

*** No improvement in chi-squared over previous iteration ***

Total time = 59.2859s	Reduced Chi^2 = 0.85382
Free parameter values:	
23.15	0.67
	68.46
	0.07955

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1
---	---

2	8		
3	23.15	+/-	0.46
4	0.67	+/-	0.03534
5	0		
6	68.46	+/-	2.109
7	0		
8	0		
9	0		
10	0		

Background parameter values:

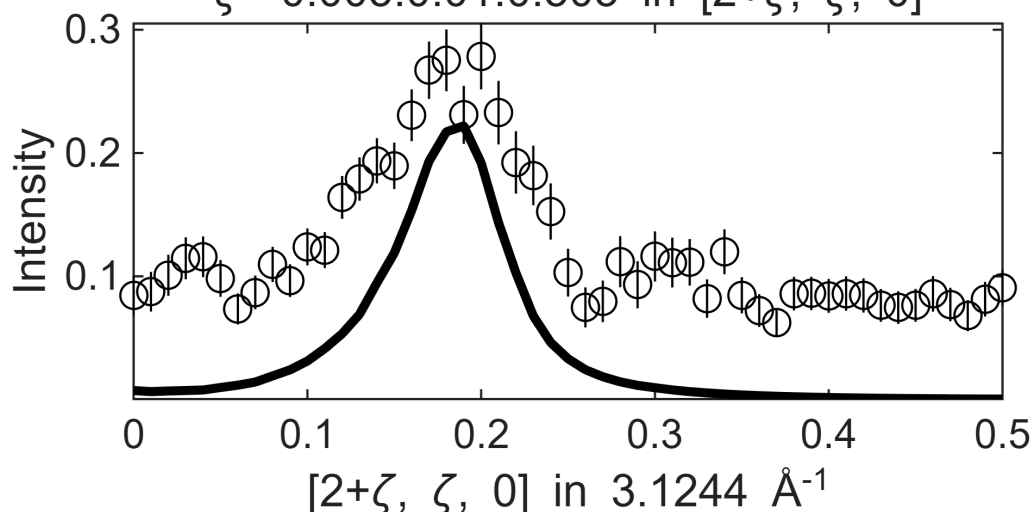
1	0.07955	+/-	0.002441
2	0		
3	0		

Covariance matrix for free parameters:

1.0000	-0.6997	0.7175	-0.1807
-0.6997	1.0000	-0.7972	-0.4486
0.7175	-0.7972	1.0000	-0.1133
-0.1807	-0.4486	-0.1133	1.0000

Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, i
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 85±5 going to 375

Beginning fit (max 20 iterations)-----

Starting point

Total time = 1.7656s Reduced Chi^2 = 14.8425

Free parameter values:

23.15	0.67	68.46	0.09232
-------	------	-------	---------

Iteration = 1

Total time = 8.6899s Reduced Chi^2 = 1.0942

Levenberg-Marquardt = 0.1

Free parameter values:

26.43	0.5423	74.68	0.05538
-------	--------	-------	---------

Iteration = 2

Total time = 15.7077s Reduced Chi^2 = 0.84673

Levenberg-Marquardt = 0.01

Free parameter values:

```

35.8      0.616      100.2      0.05075
-----
Iteration = 3
-----
Total time = 22.6513s    Reduced Chi^2 = 0.80094    Levenberg-Marquardt = 0.001
Free parameter values:
    46.03      0.6757      131.8      0.04832
-----
Iteration = 4
-----
Total time = 29.5338s    Reduced Chi^2 = 0.79048    Levenberg-Marquardt = 0.0001
Free parameter values:
    49.67      0.6864      145.4      0.04787
-----
Iteration = 5
-----
Total time = 36.5321s    Reduced Chi^2 = 0.79321    Levenberg-Marquardt = 1e-05
Free parameter values:
    49.93      0.686      146.7      0.04786
Total time = 38.2865s    Reduced Chi^2 = 0.79329    Levenberg-Marquardt = 0.0001
Free parameter values:
    49.92      0.686      146.7      0.04786
Total time = 40.0185s    Reduced Chi^2 = 0.79284    Levenberg-Marquardt = 0.001
Free parameter values:
    49.89      0.6859      146.6      0.04786
Total time = 41.7758s    Reduced Chi^2 = 0.79245    Levenberg-Marquardt = 0.1
Free parameter values:
    49.67      0.6854      145.8      0.04789
Total time = 43.5089s    Reduced Chi^2 = 0.79358    Levenberg-Marquardt = 10
Free parameter values:
    49.66      0.6863      145.5      0.04787
Total time = 45.2454s    Reduced Chi^2 = 0.79674    Levenberg-Marquardt = 1000
Free parameter values:
    49.67      0.6864      145.4      0.04787
*** No improvement in chi-squared over previous iteration ***
Total time = 45.2522s    Reduced Chi^2 = 0.79048
Free parameter values:
    49.67      0.6864      145.4      0.04787
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

Foreground parameter values:
-----
 1      1
 2      8
 3      49.67 +/- 0.8979
 4      0.6864 +/- 0.03535
 5      0
 6      145.4 +/- 4.623
 7      0
 8      0
 9      0
10      0

Background parameter values:
-----
 1      0.04787 +/- 0.002093
 2      0
 3      0

Covariance matrix for free parameters:
-----
 1.0000 -0.3686 0.6078 -0.4179

```

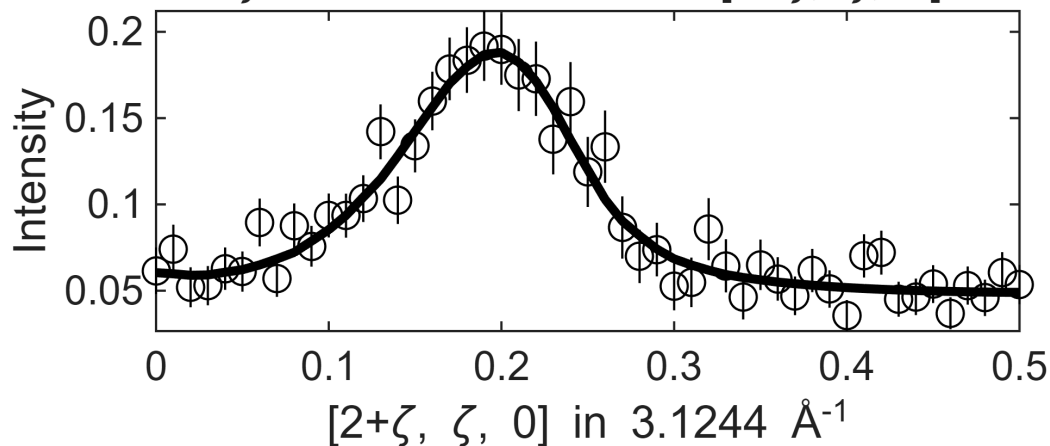
-0.3686	1.0000	-0.6390	-0.5638
0.6078	-0.6390	1.0000	-0.2228
-0.4179	-0.5638	-0.2228	1.0000

Fe_ei200_align_sym3.sqw

.035; J0=1.5e+02± 4.6; J1= 0± 0; gamma=49.67

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, ξ

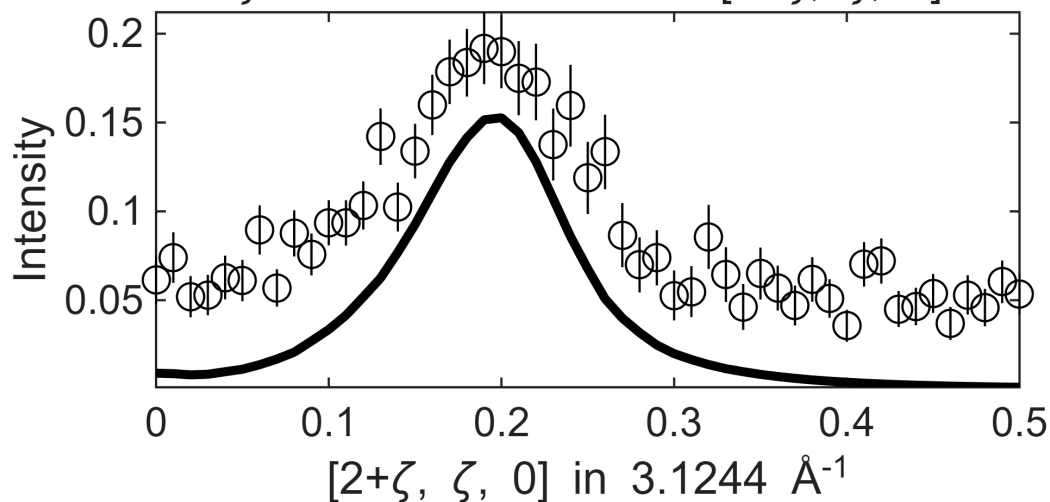
$\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, ξ

$\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 95±5 going to 375

Beginning fit (max 20 iterations)-----

Starting point

Total time = 1.869s Reduced Chi^2 = 21.0741

Free parameter values:

49.67	0.6864	145.4	0.07742
-------	--------	-------	---------

Iteration = 1

Total time = 9.2913s Reduced Chi^2 = 0.92691 Levenberg-Marquardt = 0.1

Free parameter values:

50.69	0.6615	141.3	0.03332
-------	--------	-------	---------

Iteration = 2

```

-----
Total time = 16.7665s    Reduced Chi^2 = 0.86587    Levenberg-Marquardt = 0.01
Free parameter values:
    51.75    0.6946    144.1    0.03005
-----
Iteration = 3
-----
Total time = 24.159s    Reduced Chi^2 = 0.86173    Levenberg-Marquardt = 0.001
Free parameter values:
    52.83    0.6987    147.7    0.02987
-----
Iteration = 4
-----
Total time = 31.5009s    Reduced Chi^2 = 0.86505    Levenberg-Marquardt = 0.0001
Free parameter values:
    53.21    0.6998    149    0.02982
Total time = 33.3024s    Reduced Chi^2 = 0.86498    Levenberg-Marquardt = 0.001
Free parameter values:
    53.18    0.6997    148.9    0.02982
Total time = 35.1391s    Reduced Chi^2 = 0.86456    Levenberg-Marquardt = 0.01
Free parameter values:
    53.04    0.6993    148.4    0.02984
Total time = 36.9679s    Reduced Chi^2 = 0.86543    Levenberg-Marquardt = 1
Free parameter values:
    52.86    0.6987    147.8    0.02987
Total time = 38.7723s    Reduced Chi^2 = 0.86645    Levenberg-Marquardt = 100
Free parameter values:
    52.83    0.6987    147.7    0.02987
Total time = 40.627s    Reduced Chi^2 = 0.86216    Levenberg-Marquardt = 10000
Free parameter values:
    52.83    0.6987    147.7    0.02987
*** No improvement in chi-squared over previous iteration ***
Total time = 40.6334s    Reduced Chi^2 = 0.86173
Free parameter values:
    52.83    0.6987    147.7    0.02987
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

Foreground parameter values:
-----
 1          1
 2          8
 3      52.83 +/- 1.215
 4      0.6987 +/- 0.03195
 5          0
 6      147.7 +/- 4.734
 7          0
 8          0
 9          0
10          0

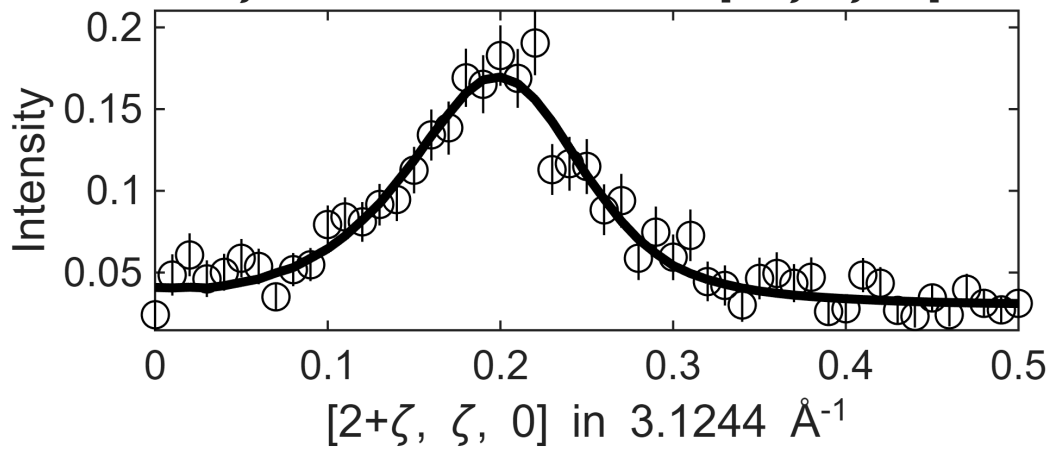
Background parameter values:
-----
 1      0.02987 +/- 0.001864
 2          0
 3          0

Covariance matrix for free parameters:
-----
 1.0000 -0.0230  0.7508 -0.7332
-0.0230  1.0000 -0.2728 -0.5865
 0.7508 -0.2728  1.0000 -0.5866
-0.7332 -0.5865 -0.5866  1.0000

```

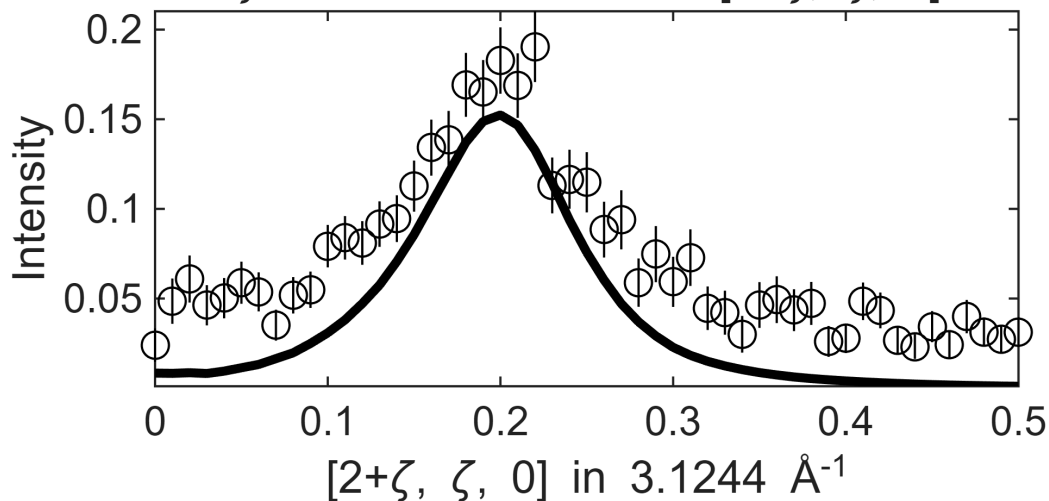
Fe_ei200_align_sym3.sqw

.032; J0=1.5e+02± 4.7; J1= 0± 0; gamma=52.83
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 9
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 9
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 105±5 going to 375

Beginning fit (max 20 iterations)-----

Starting point

Total time = 2.0263s Reduced Chi^2 = 28.3146

Free parameter values:

52.83 0.6987 147.7 0.06754

Iteration = 1

Total time = 9.9341s Reduced Chi^2 = 1.3959

Levenberg-Marquardt = 0.1

Free parameter values:

54.83 0.6229 141.4 0.02517

Iteration = 2

Total time = 17.8749s Reduced Chi^2 = 1.3008

Levenberg-Marquardt = 0.01

Free parameter values:

```

55.44      0.6605      140.4      0.0218
-----
Iteration = 3
-----
Total time = 25.7614s      Reduced Chi^2 = 1.2996      Levenberg-Marquardt = 0.001
Free parameter values:
55.88      0.6631      141.8      0.02169
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

```

Foreground parameter values:

```

-----
1          1
2          8
3          55.88 +/- 11.16
4          0.6631 +/- 0.04728
5          0
6          141.8 +/- 34.23
7          0
8          0
9          0
10         0

```

Background parameter values:

```

-----
1          0.02169 +/- 0.002385
2          0
3          0

```

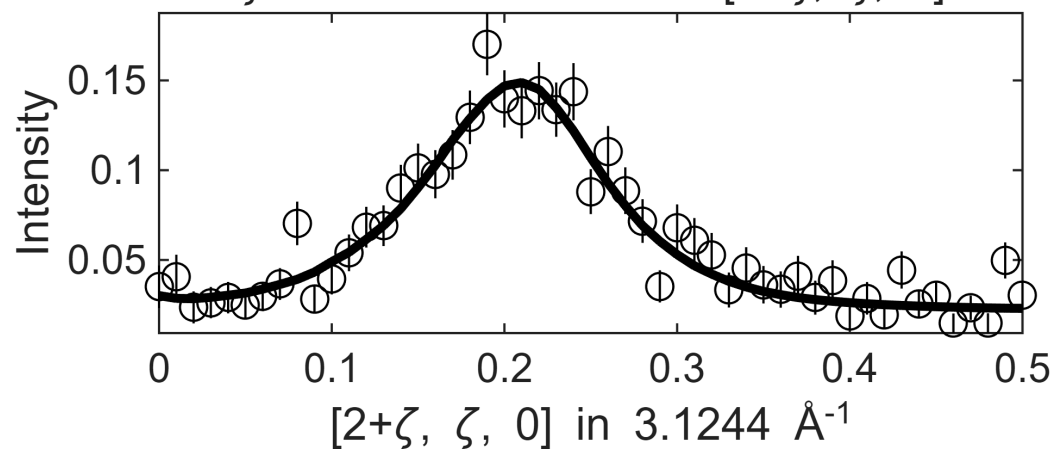
Covariance matrix for free parameters:

```

-----
1.0000    0.6553    0.9950   -0.5975
0.6553    1.0000    0.6306   -0.7389
0.9950    0.6306    1.0000   -0.5883
-0.5975   -0.7389   -0.5883    1.0000

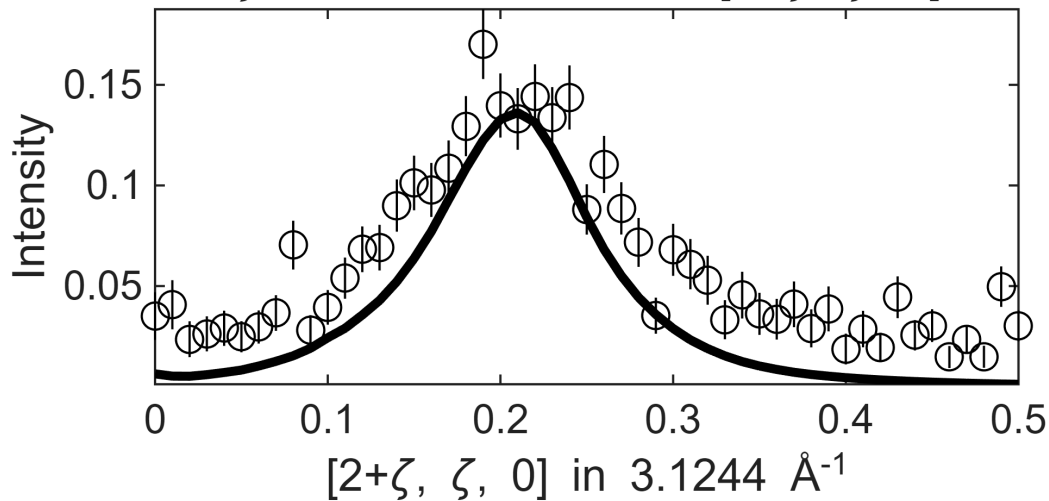
```

Fe_ei200_align_sym3.sqw
.047; J0=1.4e+02± 34; J1= 0± 0; gamma=55.88
0.1 in [2+ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, -η] , 10
ζ=-0.005:0.01:0.505 in [2+ζ, ζ, 0]



Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 10
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 115±5 going to 375

Beginning fit (max 20 iterations)-----

Starting point

 Total time = 2.3089s Reduced Chi^2 = 30.454

Free parameter values:

55.88 0.6631 141.8 0.05456

 Iteration = 1

 Total time = 11.6079s Reduced Chi^2 = 0.72574 Levenberg-Marquardt = 0.1

Free parameter values:

58.37 0.6411 135.6 0.01706

 Iteration = 2

 Total time = 20.7727s Reduced Chi^2 = 0.61106 Levenberg-Marquardt = 0.01

Free parameter values:

56.48 0.6779 129.1 0.01416

 Iteration = 3

 Total time = 30.1025s Reduced Chi^2 = 0.60773 Levenberg-Marquardt = 0.001

Free parameter values:

54.94 0.6742 125.1 0.01435

 Iteration = 4

 Total time = 39.3377s Reduced Chi^2 = 0.60749 Levenberg-Marquardt = 0.0001

Free parameter values:

54.83 0.6742 124.8 0.01437

 Fit converged:

Parameter values (free parameters with error estimates):

 Foreground parameter values:

 1 1
 2 8
 3 54.83 +/- 6.35

4	0.6742	+/-	0.02984
5	0		
6	124.8	+/-	17.17
7	0		
8	0		
9	0		
10	0		

Background parameter values:

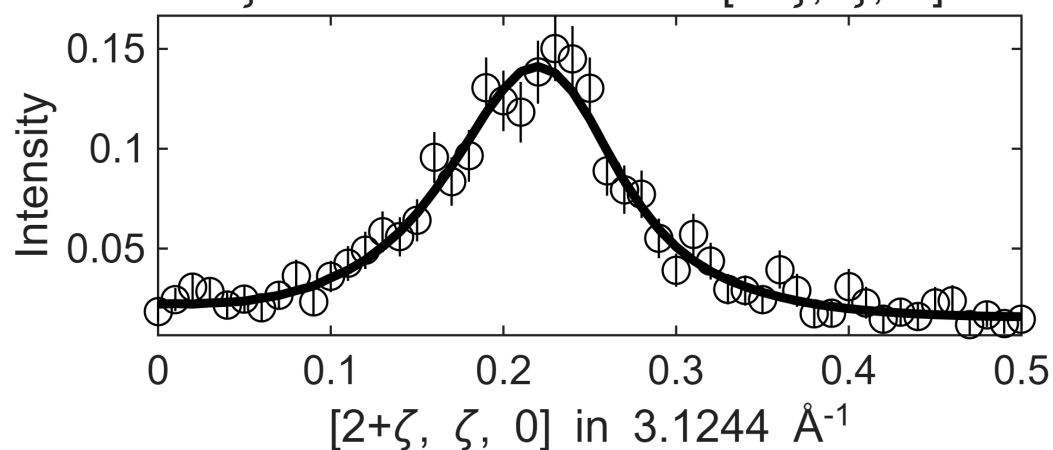
1	0.01437	+/-	0.001388
2	0		
3	0		

Covariance matrix for free parameters:

1.0000	0.6046	0.9947	-0.6472
0.6046	1.0000	0.5798	-0.7424
0.9947	0.5798	1.0000	-0.6371
-0.6472	-0.7424	-0.6371	1.0000

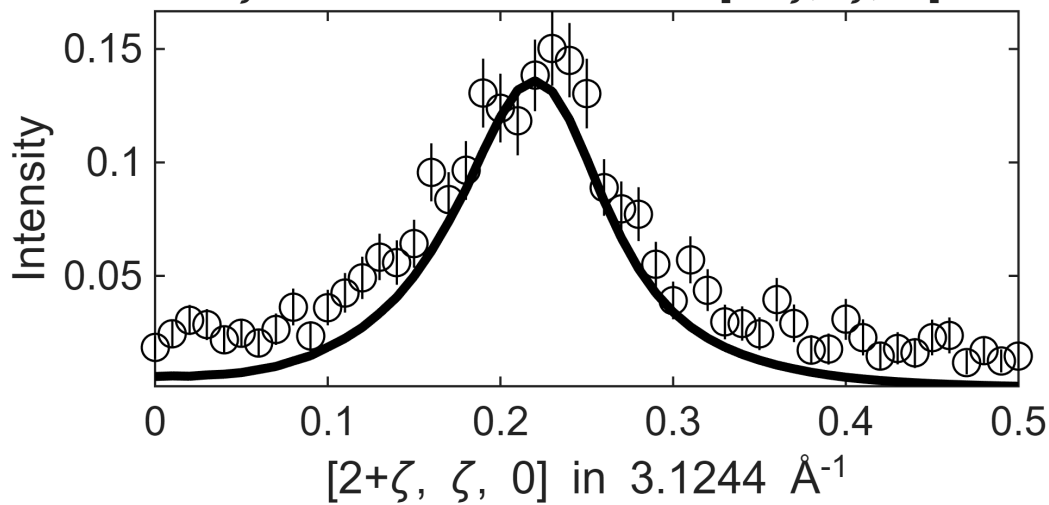
Fe_ei200_align_sym3.sqw

0.03; J0=1.2e+02± 17; J1= 0± 0; gamma=54.82°
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 1°
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $1'$
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 125±5 going to 375

Beginning fit (max 20 iterations)-----

Starting point

 Total time = 3.3791s Reduced Chi^2 = 36.6337

Free parameter values:

54.83 0.6742 124.8 0.04501

 Iteration = 1

 Total time = 17.0321s Reduced Chi^2 = 0.9508 Levenberg-Marquardt = 0.1

Free parameter values:

58.51 0.6234 122.5 0.01545

 Iteration = 2

 Total time = 31.7344s Reduced Chi^2 = 0.81191 Levenberg-Marquardt = 0.01

Free parameter values:

59.59 0.6611 122.9 0.01294

 Iteration = 3

 Total time = 46.6155s Reduced Chi^2 = 0.8123 Levenberg-Marquardt = 0.001

Free parameter values:

59.5 0.6618 122.8 0.01291

Total time = 50.3348s Reduced Chi^2 = 0.81294 Levenberg-Marquardt = 0.01

Free parameter values:

59.53 0.6619 122.9 0.01291

Total time = 54.0572s Reduced Chi^2 = 0.81055 Levenberg-Marquardt = 0.1

Free parameter values:

59.55 0.6619 123 0.01291

 Iteration = 4

 Total time = 69.0401s Reduced Chi^2 = 0.81052 Levenberg-Marquardt = 0.01

Free parameter values:

59.55 0.662 123 0.0129

 Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	59.55	+/-	7.074
4	0.662	+/-	0.03124
5	0		
6	123	+/-	17.08
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.0129	+/-	0.001259
2	0		
3	0		

Covariance matrix for free parameters:

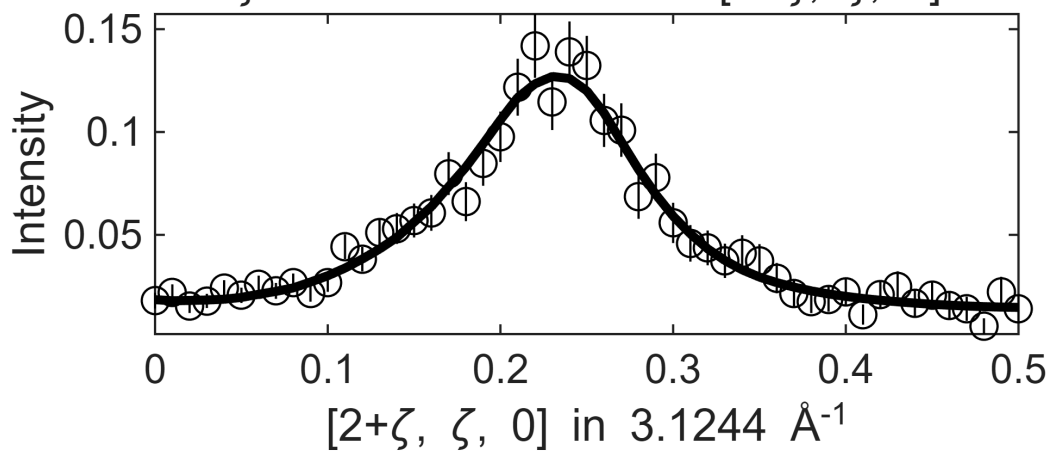
1.0000	0.5719	0.9941	-0.6712
0.5719	1.0000	0.5392	-0.7283
0.9941	0.5392	1.0000	-0.6653
-0.6712	-0.7283	-0.6653	1.0000

Fe_ei200_align_sym3.sqw

.031; J0=1.2e+02± 17; J1= 0± 0; gamma=59.55

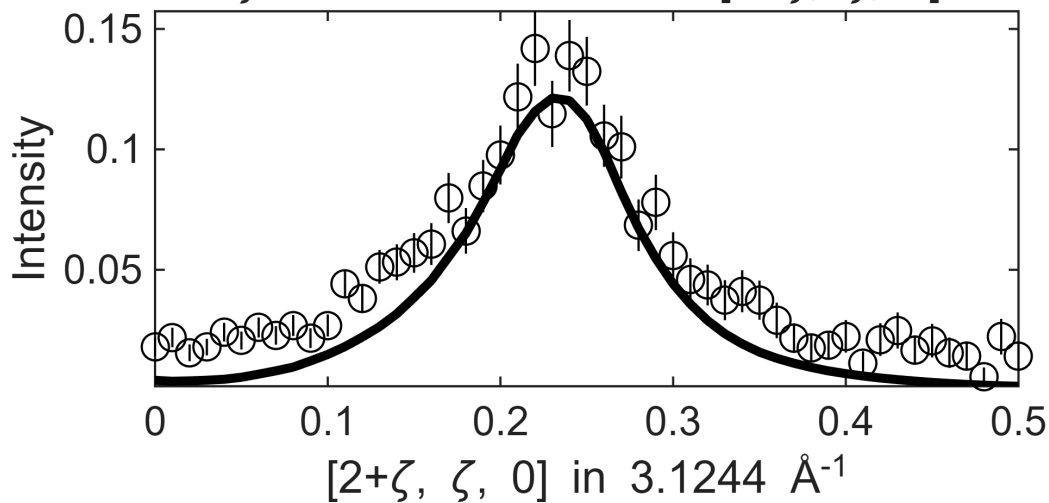
0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 1;

$\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 1;
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 135±5 going to 375

Beginning fit (max 20 iterations)-----

Starting point

 Total time = 6.6713s Reduced Chi^2 = 63.2813

Free parameter values:

59.55 0.662 123 0.0417

 Iteration = 1

 Total time = 33.4583s Reduced Chi^2 = 1.2532 Levenberg-Marquardt = 0.1

Free parameter values:

63.26 0.5487 123.9 0.0155

 Iteration = 2

 Total time = 60.2565s Reduced Chi^2 = 1.044 Levenberg-Marquardt = 0.01

Free parameter values:

68.38 0.5794 133 0.01326

 Iteration = 3

 Total time = 86.9073s Reduced Chi^2 = 1.041 Levenberg-Marquardt = 0.001

Free parameter values:

70.91 0.5844 138.9 0.01301

 Iteration = 4

 Total time = 113.7011s Reduced Chi^2 = 1.0421 Levenberg-Marquardt = 0.0001

Free parameter values:

71.42 0.5848 140.3 0.01296

Total time = 120.3661s Reduced Chi^2 = 1.0387 Levenberg-Marquardt = 0.001

Free parameter values:

71.37 0.5847 140.1 0.01297

 Iteration = 5

 Total time = 147.0364s Reduced Chi^2 = 1.0426 Levenberg-Marquardt = 0.0001

Free parameter values:

71.6 0.5854 140.7 0.01295

Total time = 153.7915s Reduced Chi^2 = 1.0412 Levenberg-Marquardt = 0.001

Free parameter values:

```

71.57      0.5853      140.6      0.01295
Total time = 160.5334s   Reduced Chi^2 = 1.0403   Levenberg-Marquardt = 0.01
Free parameter values:
71.49      0.5852      140.4      0.01296
Total time = 167.3335s   Reduced Chi^2 = 1.0421   Levenberg-Marquardt = 1
Free parameter values:
71.38      0.5848      140.2      0.01297
Total time = 174.0377s   Reduced Chi^2 = 1.0422   Levenberg-Marquardt = 100
Free parameter values:
71.37      0.5847      140.2      0.01297
Total time = 180.8422s   Reduced Chi^2 = 1.0419   Levenberg-Marquardt = 10000
Free parameter values:
71.37      0.5847      140.1      0.01297
*** No improvement in chi-squared over previous iteration ***
Total time = 180.8487s   Reduced Chi^2 = 1.0387
Free parameter values:
71.37      0.5847      140.1      0.01297

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	71.37	+/-	1.714
4	0.5847	+/-	0.02243
5	0		
6	140.1	+/-	4.61
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.01297	+/-	0.0008968
2	0		
3	0		

Covariance matrix for free parameters:

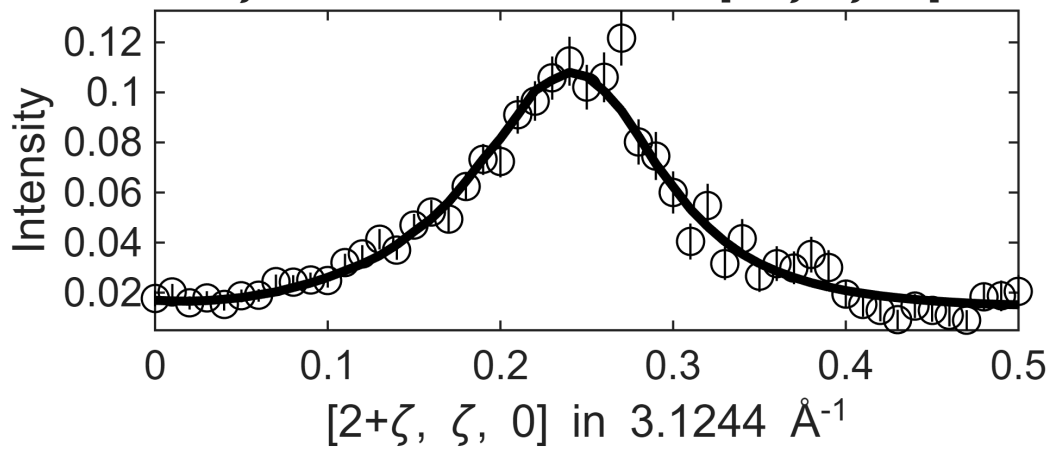
1.0000	0.2094	0.8846	-0.8805
0.2094	1.0000	-0.0468	-0.6080
0.8846	-0.0468	1.0000	-0.7509
-0.8805	-0.6080	-0.7509	1.0000

Fe_ei200_align_sym3.sqw

.022; J0=1.4e+02± 4.6; J1= 0± 0; gamma=71.36

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 1;

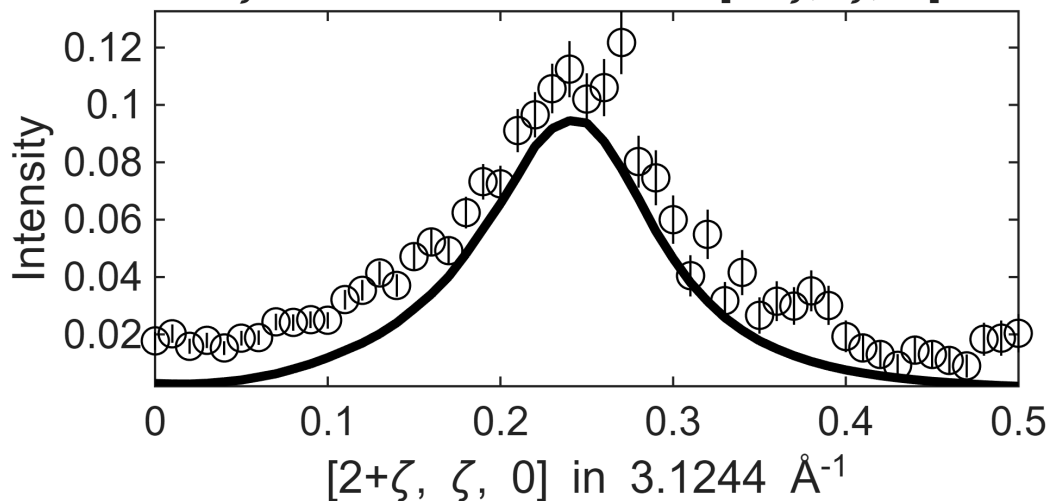
$\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 1;

$\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 145±5 going to 375

Beginning fit (max 20 iterations)-----

Starting point

Total time = 9.3459s Reduced Chi^2 = 124.3031

Free parameter values:

71.37 0.5847 140.1 0.05329

Iteration = 1

Total time = 46.2291s Reduced Chi^2 = 1.5698

Levenberg-Marquardt = 0.1

Free parameter values:

71.55 0.5419 132.1 0.01507

Iteration = 2

Total time = 80.6306s Reduced Chi^2 = 1.1164

Levenberg-Marquardt = 0.01

Free parameter values:

```

        69.3          0.5779          126.6          0.01173
-----
Iteration = 3
-----
Total time = 114.677s    Reduced Chi^2 = 1.1133    Levenberg-Marquardt = 0.001
Free parameter values:
    67.78          0.5736          123.5          0.0119
-----
Iteration = 4
-----
Total time = 148.5452s    Reduced Chi^2 = 1.1136    Levenberg-Marquardt = 0.0001
Free parameter values:
    67.46          0.5726          122.8          0.01195
Total time = 157.0208s    Reduced Chi^2 = 1.1141    Levenberg-Marquardt = 0.001
Free parameter values:
    67.49          0.5727          122.9          0.01195
Total time = 165.5485s    Reduced Chi^2 = 1.1141    Levenberg-Marquardt = 0.01
Free parameter values:
    67.61          0.5731          123.1          0.01193
Total time = 174.0956s    Reduced Chi^2 = 1.1138    Levenberg-Marquardt = 1
Free parameter values:
    67.76          0.5735          123.4          0.01191
Total time = 182.6724s    Reduced Chi^2 = 1.114    Levenberg-Marquardt = 100
Free parameter values:
    67.78          0.5736          123.5          0.0119
Total time = 191.2048s    Reduced Chi^2 = 1.1138    Levenberg-Marquardt = 10000
Free parameter values:
    67.78          0.5736          123.5          0.0119
*** No improvement in chi-squared over previous iteration ***
Total time = 191.2113s    Reduced Chi^2 = 1.1133
Free parameter values:
    67.78          0.5736          123.5          0.0119
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

Foreground parameter values:
-----
    1          1
    2          8
    3          67.78 +/- 2.115
    4          0.5736 +/- 0.01911
    5          0
    6          123.5 +/- 4.501
    7          0
    8          0
    9          0
   10          0

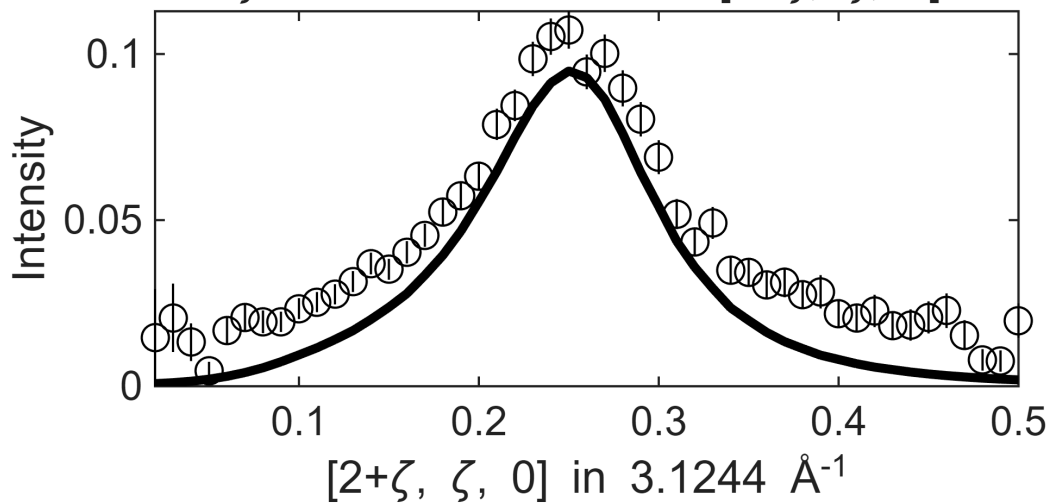
Background parameter values:
-----
    1          0.0119 +/- 0.001085
    2          0
    3          0

Covariance matrix for free parameters:
-----
    1.0000    0.6684    0.9610   -0.9684
    0.6684    1.0000    0.5629   -0.8116
    0.9610    0.5629    1.0000   -0.9353
   -0.9684   -0.8116   -0.9353    1.0000

```

Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 1°
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 155±5 going to 375

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Beginning fit (max 20 iterations)-----

Starting point

Total time = 7.6689s Reduced Chi^2 = 50.5254

Free parameter values:

67.78 0.5736 123.5 0.04094

Iteration = 1

Total time = 38.2528s Reduced Chi^2 = 1.1678 Levenberg-Marquardt = 0.1

Free parameter values:

67.34 0.5037 113 0.0185

Iteration = 2

Total time = 68.8917s Reduced Chi^2 = 0.88432 Levenberg-Marquardt = 0.01

Free parameter values:

58.68 0.5139 93.86 0.01699

Iteration = 3

Total time = 99.514s Reduced Chi^2 = 0.84624 Levenberg-Marquardt = 0.001

Free parameter values:

53.36 0.4951 84.61 0.01763

Iteration = 4

Total time = 130.073s Reduced Chi^2 = 0.84423 Levenberg-Marquardt = 0.0001

Free parameter values:

52.84 0.4931 83.69 0.01768

Iteration = 5

Total time = 160.5749s Reduced Chi^2 = 0.84597 Levenberg-Marquardt = 1e-05

Free parameter values:

52.82 0.4932 83.66 0.01768

Total time = 168.2754s Reduced Chi^2 = 0.84582 Levenberg-Marquardt = 0.0001

Free parameter values:

52.82 0.4932 83.66 0.01768

```

Total time = 176.0113s    Reduced Chi^2 = 0.84451    Levenberg-Marquardt = 0.001
Free parameter values:
    52.82    0.4932    83.67    0.01768
Total time = 183.7598s    Reduced Chi^2 = 0.84527    Levenberg-Marquardt = 0.1
Free parameter values:
    52.83    0.4932    83.69    0.01768
Total time = 191.4422s    Reduced Chi^2 = 0.84579    Levenberg-Marquardt = 10
Free parameter values:
    52.84    0.4931    83.69    0.01768
Total time = 199.1125s    Reduced Chi^2 = 0.84513    Levenberg-Marquardt = 1000
Free parameter values:
    52.84    0.4931    83.69    0.01768
*** No improvement in chi-squared over previous iteration ***
Total time = 199.1191s    Reduced Chi^2 = 0.84423
Free parameter values:
    52.84    0.4931    83.69    0.01768

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	52.84	+/-	1.11
4	0.4931	+/-	0.01619
5	0		
6	83.69	+/-	1.53
7	0		
8	0		
9	0		
10	0		

Background parameter values:

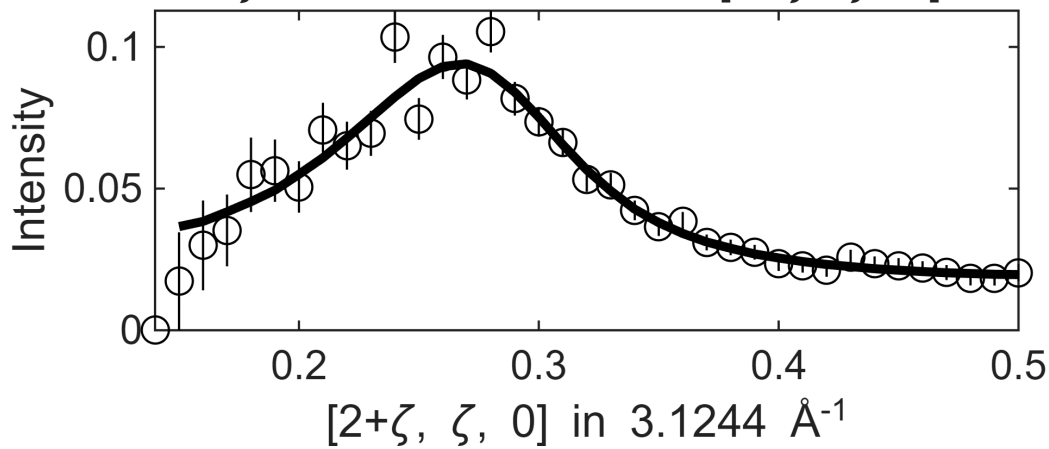
1	0.01768	+/-	0.0008228
2	0		
3	0		

Covariance matrix for free parameters:

1.0000	-0.0939	0.8671	-0.6898
-0.0939	1.0000	-0.1521	-0.6243
0.8671	-0.1521	1.0000	-0.6473
-0.6898	-0.6243	-0.6473	1.0000

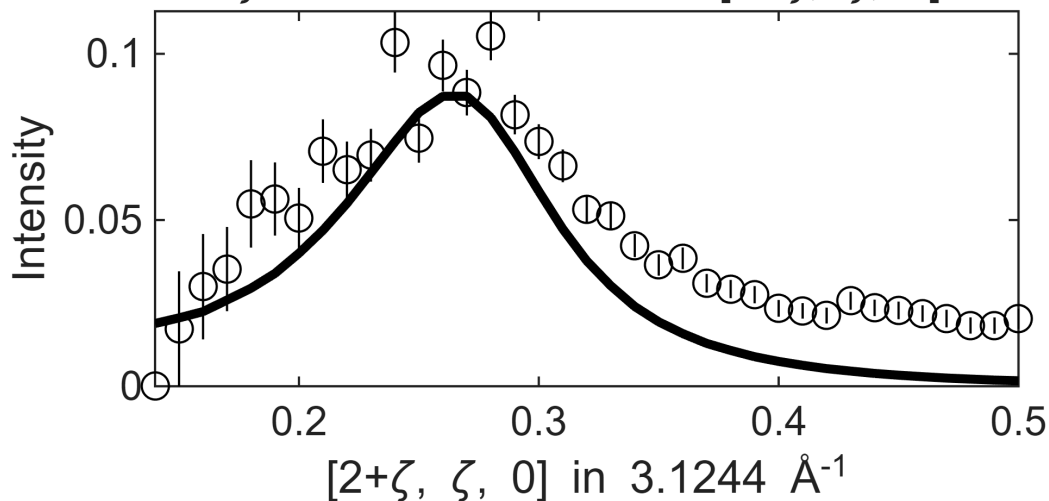
Fe_ei200_align_sym3.sqw

± 0.016 ; $J_0 = 84 \pm 1.5$; $J_1 = 0 \pm 0$; $\gamma = 52.8396$
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 1
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 1
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 165±5 going to 375

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Beginning fit (max 20 iterations)-----

Starting point

 Total time = 1.2334s Reduced Chi^2 = 1.5769

Free parameter values:

52.84 0.4931 83.69 0.02595

 Iteration = 1

 Total time = 4.3412s Reduced Chi^2 = 0.79534 Levenberg-Marquardt = 0.1

Free parameter values:

51.57 0.7541 76.26 0.01826

 Iteration = 2

Total time = 6.914s	Reduced Chi^2 = 0.84334	Levenberg-Marquardt = 0.01
Free parameter values:		
45.01	1.43	80.35
0.01627		
Total time = 7.5398s	Reduced Chi^2 = 0.77952	Levenberg-Marquardt = 0.1
Free parameter values:		
49.71	0.9744	76.98
0.01685		

Iteration = 3

Total time = 10.157s	Reduced Chi^2 = 0.82791	Levenberg-Marquardt = 0.01
Free parameter values:		
42.85	1.713	78.45
0.01613		
Total time = 10.8236s	Reduced Chi^2 = 0.78024	Levenberg-Marquardt = 0.1
Free parameter values:		
47.69	1.219	77.12
0.01649		
Total time = 11.4455s	Reduced Chi^2 = 0.77885	Levenberg-Marquardt = 1
Free parameter values:		
49.15	1.058	76.85
0.01672		

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	49.15	+/-	3743
4	1.058	+/-	163.7
5	0		
6	76.85	+/-	4180
7	0		
8	0		
9	0		
10	0		

Background parameter values:

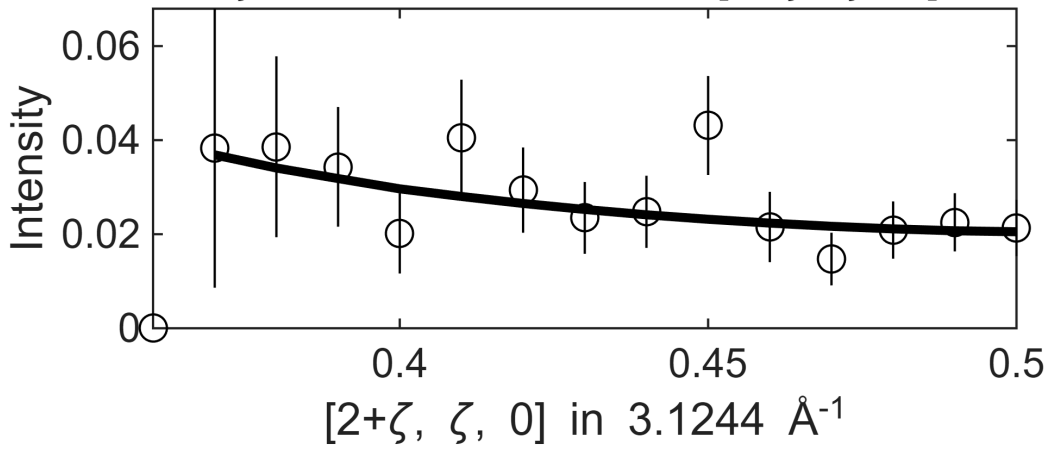
1	0.01672	+/-	0.03227
2	0		
3	0		

Covariance matrix for free parameters:

1.0000	-0.9957	0.9971	0.8968
-0.9957	1.0000	-0.9859	-0.9324
0.9971	-0.9859	1.0000	0.8623
0.8968	-0.9324	0.8623	1.0000

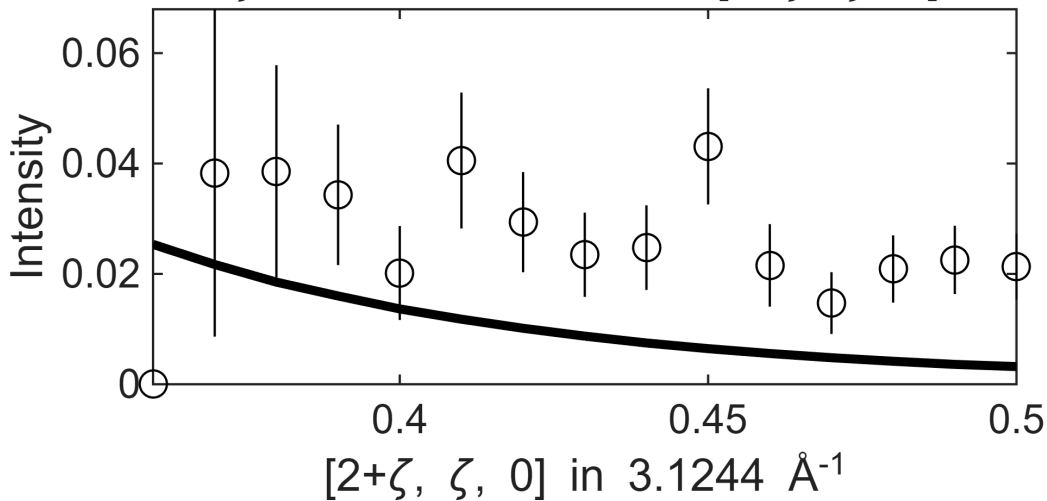
Fe_ei200_align_sym3.sqw

1e+02; J0= 77±4.2e+03; J1= 0± 0; gamma=49.1
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 10
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 10
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

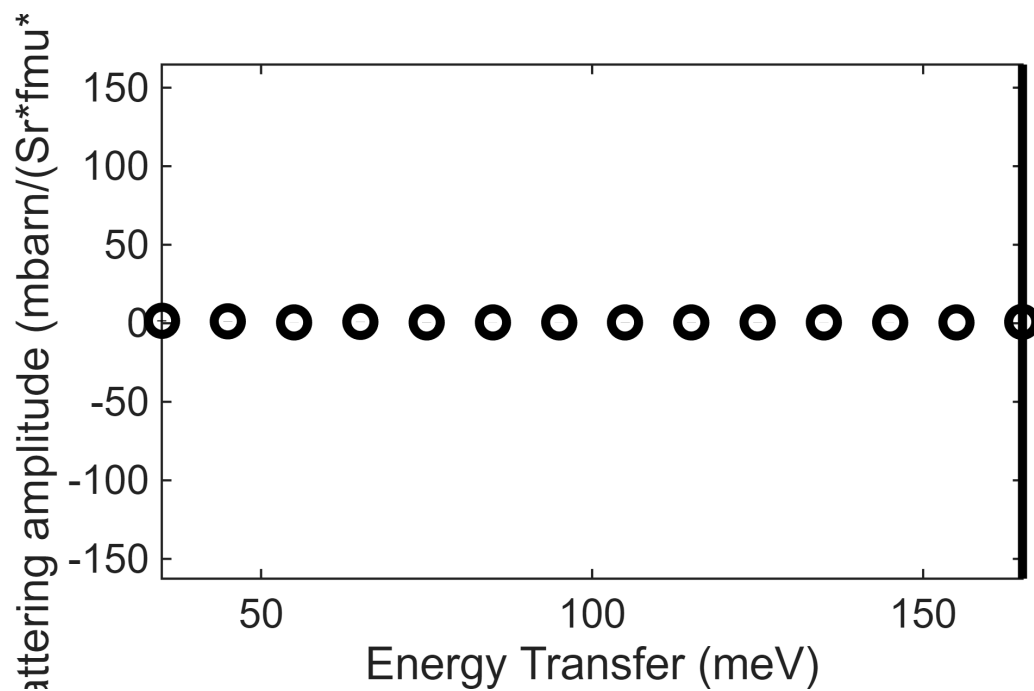


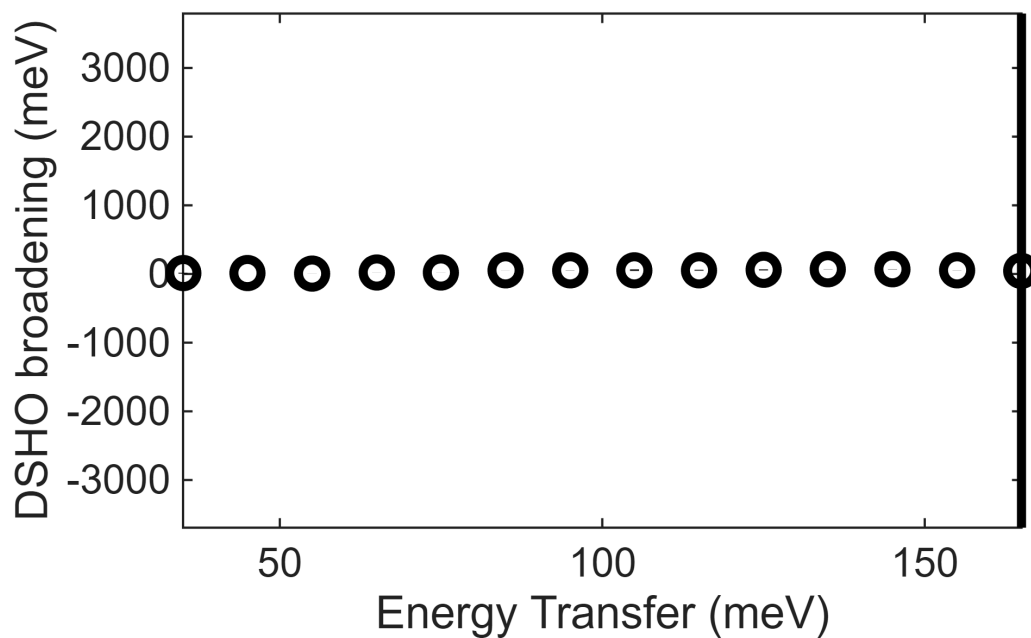
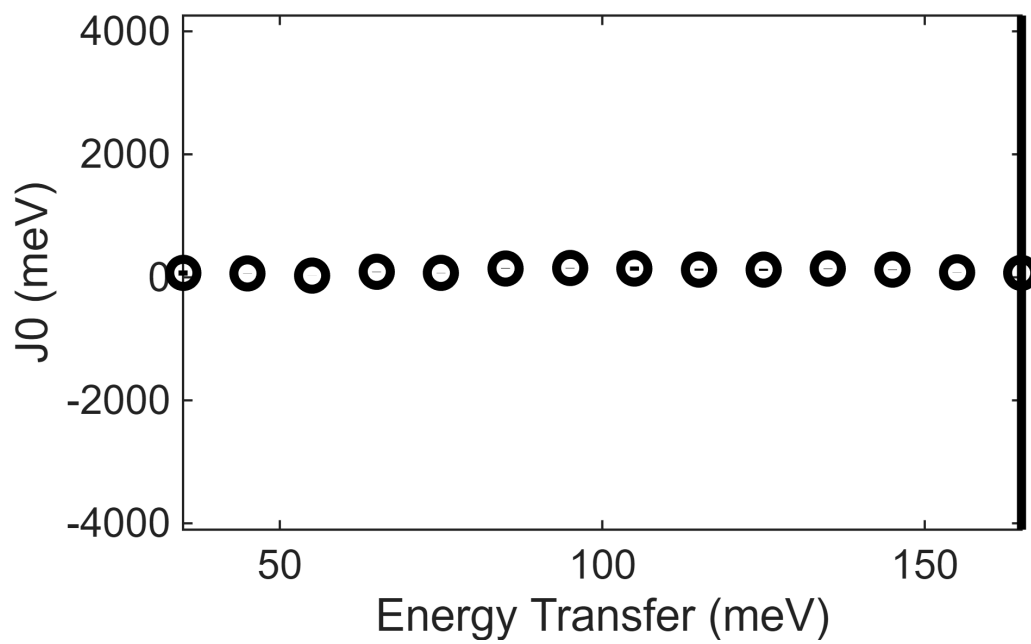
```
*****
**** En = 175±5 going to 375
*****
**** En = 185±5 going to 375
*****
**** En = 195±5 going to 375
*****
**** En = 205±5 going to 375
*****
**** En = 215±5 going to 375
*****
**** En = 225±5 going to 375
*****
**** En = 235±5 going to 375
*****
```

```

*****
**** En = 245±5 going to 375
*****
*****
**** En = 255±5 going to 375
*****
*****
**** En = 265±5 going to 375
*****
*****
**** En = 275±5 going to 375
*****
*****
**** En = 285±5 going to 375
*****
*****
**** En = 295±5 going to 375
*****
*****
**** En = 305±5 going to 375
*****
*****
**** En = 315±5 going to 375
*****
*****
**** En = 325±5 going to 375
*****
*****
**** En = 335±5 going to 375
*****
*****
**** En = 345±5 going to 375
*****
*****
**** En = 355±5 going to 375
*****
*****
**** En = 365±5 going to 375
*****
*****
**** En = 375±5 going to 375
*****

```





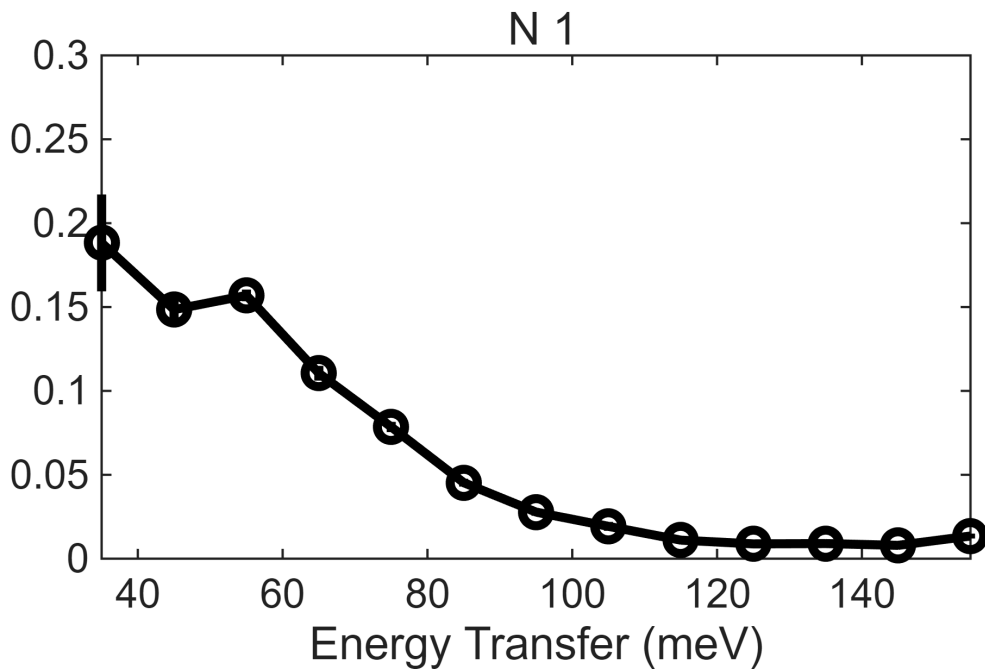
```

c1 = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [c1{:}];

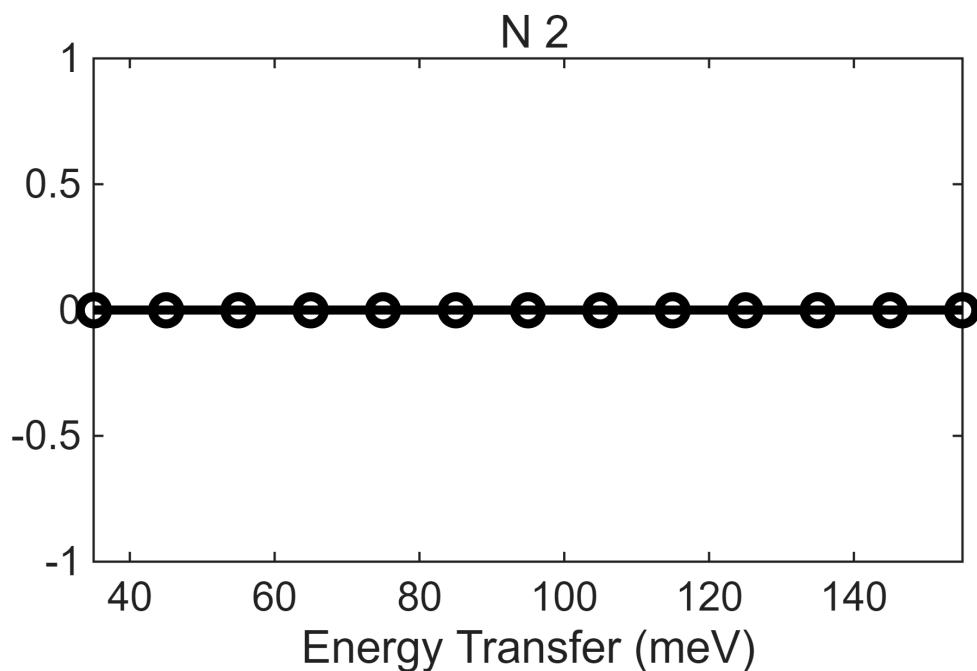
beep
beep
beep

all_fit_par = NG_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);ly 0 0.3; keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
```

```
%end
```

```
fit_src_struct = NG_FFCutsFits;  
save(CutFitFileNG, "fit_src_struct", '-v7.3')
```